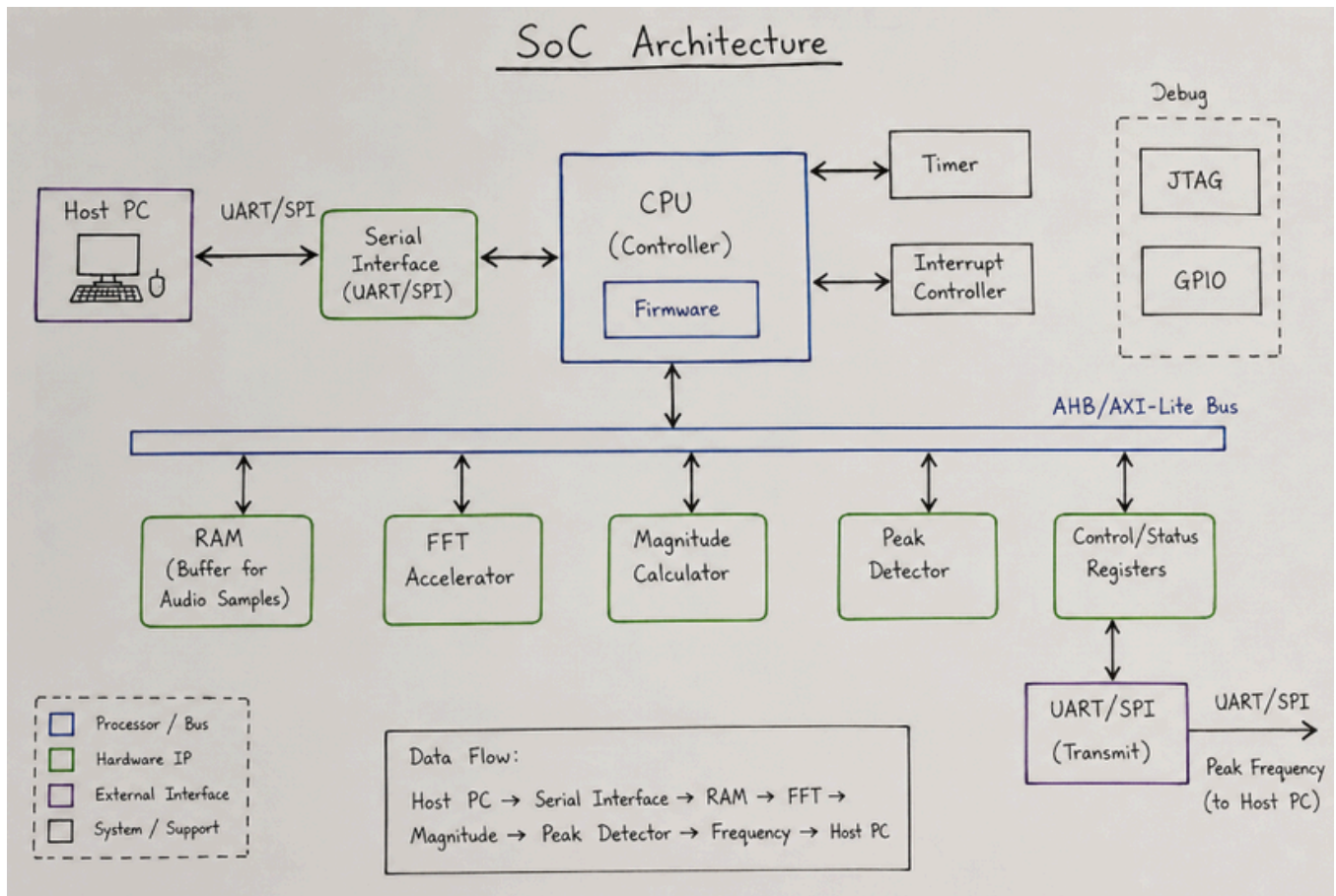


SoC Lab 1

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SOC Architecture



Hardware Software Partitioning

- **Hardware:**
 - UART/SPI Controller
 - RAM
 - FFT Computation
 - Magnitude Calculation
 - Peak Detector
 - Status Registers
- **Software:**
 - System Initialization
 - Data Reception Control
 - FFT Accelerator Control
 - Peak Bin to Frequency Conversion
 - Result Transmission
 - Overall System Control

Pseudocode

Start

Initialize UART/SPI

Repeat

 Receive audio samples from the PC

 Store samples in memory

 Start FFT

 Wait until FFT is finished

 Calculate the magnitude spectrum

 Find the peak frequency bin

 Convert the peak bin to the actual frequency

 Send the frequency back to the PC through UART/SPI

End Repeat